Department of Economics
University of Arkansas
Fall 2013

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Tue/Thu 1.30PM-2.30PM

ECON 5613 - Econometrics I
9.30 - 10.45AM Tue/Thu, WCOB336

Course Description

This course offers a rigorous, but intuitive, introduction to econometrics that provides a foundation for users of econometrics to begin engaging in applied research and to take more advanced coursework. A basic understanding of probability is assumed, as is an understanding of linear (matrix) algebra as the latter serves as a convenient short-hand notation for presenting the concepts and methods treated in the course. The course provides the theoretical background necessary to understand how to correctly apply, test, and interpret econometric techniques.

Textbooks

The main textbook for the course is Greene's *Econometric analysis*, 7th ed edition (2012). Other excellent graduate-level textbooks include Wooldridge (2010), Cameron and Trivedi (2005), and Angrist and Pischke (2009). Meanwhile, Wooldridge (2012); Murnane and Willett (2011) (in education) and Kennedy (2008) are useful supplementary textbooks that offer intuitive and accessible explanations of topics covered in the course.

William H. Greene. *Econometric analysis*. Prentice Hall, Boston, 7th ed edition, 2012. ISBN 0131395386.

Jeffrey M. Wooldridge. *Econometric analysis of cross section and panel data*. MIT Press, Cambridge, Mass, 2nd ed edition, 2010. ISBN 9780262232586.

Adrian Colin Cameron and Pravin K. Trivedi. *Microeconometrics: methods and applications*. Cambridge University Press, Cambridge ; New York, 2005. ISBN 9780521848053.

Joshua David Angrist and Jorn-Steffen Pischke. *Mostly harmless econometrics: an empiricist's companion*. Princeton University Press, Princeton, 2009. ISBN 9780691120348.

Jeffrey M. Wooldridge. *Introductory econometrics: a modern approach*. South-Western Cengage Learning, Mason, OH, 5th ed edition, 2012. ISBN 9781111531041.

Richard J. Murnane and John B. Willett. *Methods matter: improving causal inference in educational and social science research*. Oxford University Press, Oxford ; New York, 2011. ISBN 9780199753864.

Peter Kennedy. *A guide to econometrics*. Blackwell Pub, Malden, MA, 6th ed edition, 2008. ISBN 9781405182584.

Tentative Schedule

Week	Topic	Readings
1	Introduction to Econometrics: Methods, Data, and Applications	Wooldridge (2012, Ch. 1-2)*, Greene (2012, Ch. 1), Cameron and Trivedi (2005, Ch. 2-3)
2-4	Multiple Regression, the Linear Statistical Model, Tests of Hypothesis	Greene (2012, Ch. 2-5)*, Wooldridge (2010, Ch. 4.1) Cameron and Trivedi (2005, Ch. 4.1-4.4)
5	Properties of Least Squares	Greene (2012, Ch. 4.3-4.4)*, Wooldridge (2010, Ch. 3, 4.2.1, 4.2.2)
6-7	Heteroskedasticity, Autocorrelation, and Robust Variance Estimation	Greene (2012, Ch. 9.1-9.2)*, Wooldridge (2010, Ch. 4.2.3)
8	**Midterm Exam (October 17th)**	
9-11	Generalized Least Squares, Heteroskedasticity and Serial Correlation	Greene (2012, Ch. 9.3-9.5, 20.5, 20.7)*, Wooldridge (2010, Ch. 7)
11-13	Nonorthogonality of Regressors and Errors: Correlation Between Regressors and Errors, Errors in Variables, Instrumental Variables and Specification Tests	Greene (2012, Ch. 8)*, Wooldridge (2010, Ch. 4.3-4.4, 5) Cameron and Trivedi (2005, Ch. 4.1-4.4)
14	Panel Data	Greene (2012, Ch. 11)*, Wooldridge (2010, Ch. 10) Cameron and Trivedi (2005, Ch. 21)
15-16	Nonlinear Specifications, Maximum Likelihood Estimation	Greene (2012, Ch. 14)*, Wooldridge (2010, Ch. 12.9-12.10, 13.1-13.7) Cameron and Trivedi (2005, Ch. 5)

Note: *Required reading

Grading Policy

Grading for the course will be based on problem sets (20%), the midterm (35%) and the final exam (45%). Letter grades (without the plus/minus option) will be assigned based on the total score. All students are expected to complete all problem sets and exams. There will be no provisions for extra work for extra credit.

The problem sets will be assigned approximately every other Tuesday and is due the following Tuesday. Students are encouraged to work in groups on problem sets, but it is in your interest to make a best effort to solve each problem. Obviously, each student must hand in his or her own set of answers. Completed problem sets will receive a full credit, a half credit or no credit based on quality. Missing or late problem sets will not be graded.

Problem sets may involve utilizing mathematical (e.g, Matlab) or statistical software (e.g, Stata). Please read each problem set carefully regarding the types of outputs that you need to

submit for these problems. These softwares should be available on the Walton Server. A session will be held to familiarize students with these softwares.

The midterm exam will be held during class time on October 17th. **The final exam** will be on December 19th, between 8.00-10.00AM. Students will be permitted to bring one page of notes to the first exam and three pages of notes to the final exam

Academic Dishonesty

Students are expected to adhere to the Universitys policy regarding academic honesty. It is explained in the Catalog of Studies, and can be found online at: http://catalogofstudies.uark.edu/1213-Undergraduate_Catalog-1.pdf

Accommodations for Students with Disabilities

University of Arkansas Academic Policy requires that students with disabilities are provided reasonable accommodations to ensure their equal access to course content. If you have a documented disability and require accommodations, please contact me privately at the beginning of the semester to make arrangements for necessary classroom adjustments. Please note, you must first verify your eligibility for these through the Center for Educational Access (contact 479-575-3104 or visit http://cea.uark.edu for more information on registration procedures).

Inclement Weather Policy

In case of severe weather, a notification will be sent out via Blackboard regarding cancellation of any class. Further instructions regarding the make-up of missed class will also be posted on the Blackboard site.